

a) Fill the following timing table for executing program A in this processor. (2 marks)

Cycle	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
Instruction												

b) Based on this processor, in which stage (step) control lines are created? (1 marks)

c) Determine control line values that created in cycle 3, and for which instruction? (2.5 marks)

Instruction is

RegDst	ALU OP1	ALU OP2	ALUSrc	Branch	MemRead	MemWrite	Regwrite	MemtoReg

d) Determine control line values that created in cycle 4, and for which instruction? (2.5 marks)

Instruction is

RegDst	ALU OP1	ALU OP2	ALUSrc	Branch	MemRead	MemWrite	Regwrite	MemtoReg

e) One designer suggests to remove the big registers (A, B, C and D) from the design to decrease coast. Is this suggestion valid or not? Why? (2 marks)

- Q5) Identify three types of hazards encountered in instruction pipelining in computing systems. suggest techniques to mitigate each hazard type (3 marks)

Q6) Modern processors have significantly outpaced memory speeds, leading to a performance bottleneck in many computing systems. Imagine you are designing a new processor architecture (3 marks)

- a) What technique or technology could you implement to mitigate the memory bottleneck and ensure the processor's speed advantage is effectively utilized?(1 marks)

- b) One designer suggests to increase the size of cache to be equal RAM size. Is this suggestion valid or not? Why? (1 marks)

- c) Based on Principle of Locality, give me one example of each temporal locality and spatial locality (1marks)

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instruction memory

Register file

Memory data

a) Fill the marks)

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Q1) For the loop:

for ($i = 0$; $i < 500$; $i = i + 2$) { $y[i] = y[i] + k + z$; }

The MIPS assembly code for this loop is given as:

```
lw $4, 0($1)    ; load k into $4
lw $5, 0($3)    ; load z into $5
addi $6, $2, 2000 ; $6 = address of last element + 4
loop: lw $7, 0($2) ; load y[i] into $7
      add $8, $7, $4 ; $8 = y[i] + k
      add $9, $8, $5 ; $9 = y[i] + k + z
      sw $9, 0($2) ; store computed y[i]
      addi $2, $2, 8 ; increment $2 to point to next y[ ] element (i + 2)
      bne $6, $2, loop ; last loop iteration reached
```

The MIPS code is executed on a CPU that runs at 500 MHz and has the following instruction type CPIs:

Instruction Type	CPI
ALU	3
load	5
Store	7
branch	4

Based on this information, calculate (12 marks):

- a) Fraction of total instructions executed for each instruction type. (4marks)

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b) Total number of CPU cycles. (1mark)

c) Average CPI. (1mark)

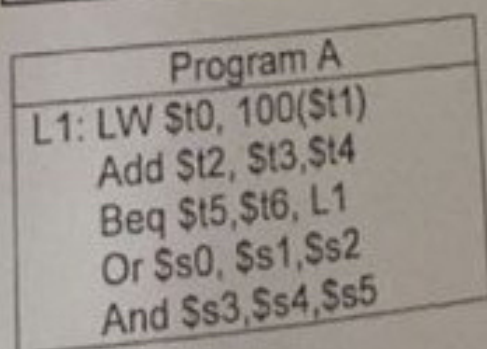
d) Fraction of total execution time for each instruction type. (4marks)

e) Total execution time for the loop. (2 marks)



← Reply

- Q5) The following is the MIPS processor use pipelining and snapshot of program A. (13 marks)



Q1) Consider two different implementations of the same instruction set architecture. There are four classes of instructions, A, B, C, and D. The clock rate and CPI of each implementation are given in the following table.

Processor	Clock Rate	CPI Class A	CPI Class B	CPI Class C	CPI Class D
Processor 1	5GHZ	5	3	3	4
Processor 2	3GHZ	4	5	5	3

Based on this information, calculate (12 marks):

- a) Given a program with 10^6 instructions divided into classes as follows: 20% class A, 10% class B, 40% class C, and 30% class D, find the execution time for processor 1 and processor 2? (4marks)

time for processor 1 =

time for processor 2 =

time to determine
miss or hit + time to access
the cache

time to bring the block from
lower level to upper

SB → SAME
SNA

Q2) Consider the following C++ code snippet:

(4 marks)

```
for (i = 0; i < 100; i = i + 1) {
    if (A[i] < 10)
        sum = sum + A[i];
}
```

Assume the following:

- A is an array of integers.
- The base address of the array A is stored in the \$t1 register.
- The index variable i is stored in the \$t4 register.
- The variable sum is stored in the \$t3 register.

Write the MIPS assembly code to implement the given C++ code.

Q6) the following is a snapshot of algorithm used with pipeline processor (3 marks)

```
if (ID/EX.MemRead and  
((ID/EX.RegRt = IF/ID.RegRs) or  
(ID/EX.RegRt = IF/ID.RegRt)))  
then stall the pipeline
```

- a) This algorithm is used for _____ (1 marks)
- b) Explain this algorithm in details (2 marks)

Q2) What is "overflow" in the context of a computer processor, and when does it occur during arithmetic operations, give me four examples? (6 marks)